

# The Foundation of FFGFT in Four Layers

Fundamental Equation, Electromagnetic Sector, Scaling Parameter and Algebraic  
Confirmations

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# **The Foundation of FFGFT in Four Layers**

## Abstract

FFGFT rests on four layers of different epistemic character. Layer 1 (the fundamental equation and the principle of relativity) and Layer 2 (the electromagnetic sector with  $\alpha = 1$ ) are parameter-free in natural units and generate the structure. Layer 3 ( $\xi$  as scaling parameter) translates the structure into measurable quantities; it is the only free parameter. Layer 4 (algebraic confirmations: Galois fields, Hodge theory, representation theory) demonstrates the coherence of the system against known mathematics but adds no new inputs. Translation into SI units and comparison with the Standard Model are communication work for established systems, not structural inputs. The cosmic sector ( $H_0/\Lambda$ ) is bracketed because the Standard Model derives it equally little.

## List of symbols

The table explains symbols appearing in more than one section. Quantities used only locally are introduced there.

| Symbol                                     | Meaning                                                 |
|--------------------------------------------|---------------------------------------------------------|
| <i>Fundamental equation</i>                |                                                         |
| $\tilde{T}$                                | intrinsic time scale of a system                        |
| $m$                                        | mass (in natural units = energy)                        |
| $\tilde{T} \cdot m = 1$                    | fundamental equation of FFGFT                           |
| <i>Geometry</i>                            |                                                         |
| $T^4$                                      | four-dimensional torus (compact carrier)                |
| $D_4$                                      | root lattice in $\mathbb{R}^4$ ; kissing number 24      |
| $\mathbb{Z}_3$                             | identification group; triality on $D_4$                 |
| $T^4/\mathbb{Z}_3$                         | orbifold; nine fixed points                             |
| <i>Parameters and fields</i>               |                                                         |
| $\xi = 4/30\,000$                          | dimensionless scaling parameter                         |
| $\alpha$                                   | fine-structure constant; in FFGFT units $\alpha = 1$    |
| $GF(3^k)$                                  | finite field of characteristic 3; mode classification   |
| $\varphi = (1+\sqrt{5})/2$                 | golden ratio; primitive element of $GF(9)^*$            |
| $\theta = 2/9$                             | output of the $\mathbb{Z}_3$ -circulant diagonalisation |
| <i>Field equations</i>                     |                                                         |
| $F^{\mu\nu}$                               | electromagnetic field strength tensor                   |
| $j^\nu$                                    | four-current                                            |
| $\not{\partial} = \gamma^\mu \partial_\mu$ | Feynman slash notation                                  |
| $\hat{H}$                                  | Hamiltonian                                             |
| <i>Operations</i>                          |                                                         |
| Type I                                     | duality decompactification; information-preserving      |
| Type II                                    | geometric projection; lossy                             |
| Type III                                   | representational translation; bijective                 |

## On the intention

FFGFT does not make predictions in the sense of asserting something contentually new. It rewrites known relations so that hidden structure becomes visible. Experimental confirmation is not the goal — the reformulated equations are identical to the established ones and therefore trivially correct. What is interesting is what the rewriting reveals: that  $H_0$  follows from the geometry, that  $a_0 = cH_0/(2\pi)$  is not a coincidence, that  $\theta = 2/9$  was not an input. Had the cosmic sector been unknown, the geometry would have supplied it — one could then measure whether it agrees. That it is already known does not make the statement false, but it makes testability harder, because one can never be fully certain whether an exponent was found or chosen.

Confirmations do not succeed — that is not a deficiency but the consequence of the method. Whoever seeks confirmation has misunderstood the intention.

## 1 Layer 1: the fundamental equation

In natural units ( $\hbar = c = 1$ ) the only fundamental equation of FFGFT is

$$\tilde{T} \cdot m = 1.$$

It states: time scale and mass are reciprocal representations of the same object. Combined with the principle of relativity (Lorentz invariance, causal structure, action principle) this implies:

- A compact carrier  $T^4$  as the smallest structure compatible with  $\tilde{T} \cdot m = 1$ , Lorentz invariance and three spatial dimensions **[POSIT]**.
- $\mathbb{Z}_3$  as the most economical identification with three generations **[POSIT]**.
- $D_4$  as the densest lattice in four dimensions with  $\mathbb{Z}_3$  triality **[POSIT]**.

These three posits are motivations, not derivations; they are vindicated by internal coherence (Dok. 248).

### Three levels of description

It is important to distinguish what belongs to the geometry of the torus and what is a translation into other mathematical languages (Dok. 270/330):

1. **Torus geometry (Level 1)** — the physics itself.  $T^4/\mathbb{Z}_3$  on the  $D_4$  lattice, modes as winding numbers, fixed points, ratios. Static, in  $\mathbb{Q}$  and algebraic fields.
2. **Galois fields (Level 2)** — the algebraic classification of the modes.  $GF(9)$ ,  $GF(27)$ ,  $GF(729)$  describe which mode orders can exist; the GF tower is the algebraic language of Level 1, not a separate level of reality. Bijective translation (Type III, Dok. 270): no information loss, no new content.
3. **Matrix/Hilbert space (Level 3)** — the QM language.  $T^4 \leftrightarrow \mathcal{H}$  via Fourier on the torus; likewise Type III, bijective. Enables connection to Schrödinger, Dirac, product states — but adds no physics. The Lagrangian and field equations belong to this level as imported vocabulary **[E]** (R120), to keep the theory compatible with established frameworks.

All three levels describe the same static snapshot in different languages. What changes is the language, not the content.

### Static picture — snapshot

The theory describes a static structure: a snapshot. Time is a coordinate of the torus  $T^4$ , not a distinguished evolution parameter;  $\tilde{T} \cdot m = 1$  is a property of each state, not a propagation law. What appears as a temporal evolution is the transition between snapshots — which the theory does not describe in Layers 1–3. The position-dependent time field  $T(x, t) \equiv 1/m(x, t)$  is the snapshot relation at point  $x$ : when the mass  $m(x, t)$  varies in a gravitational potential,  $T(x, t)$  varies with it necessarily — not because  $T$  is an independent dynamical field, but because  $T = 1/m$  holds at every instant. The “extended” Schrödinger equation with  $T(x, t)$  formulated in the early documents is therefore a rewriting of the mass-position-dependent coupling to the gravitational potential, not a new degree of freedom.

### Ratio-based computation and exact rational arithmetic

Since  $\tilde{T} \cdot m = 1$  in natural units formulates all physical statements as ratios — mass to mass, time to time, length to length — the fundamental calculations remain in  $\mathbb{Q}$  or finite algebraic extensions such as  $\mathbb{Q}(\sqrt{5})$ . No floating-point operations are required as long as no SI anchor is inserted. The FFGFT verification scripts therefore use exact rational arithmetic (SymPy Rational, Python’s Fraction); replay identity — bit-identical reproducibility of every step

— is the natural test criterion. Rounded approximations arise only when translating into SI units (Layer 3).

**What already follows.** From  $\tilde{T} \cdot m = 1$  alone follow time dilation as a mass effect, red-shift as path lengthening in the intrinsic time field and the basic structure of gravitation (Dok. 078/270). The pattern of the mass ladder follows from the geometry of  $T^4$ ;  $\theta = 2/9$  is the output of the  $\mathbb{Z}_3$ -circulant diagonalisation, not an input (Dok. 230/231, “found, not sought”).

## 2 Layer 2: the electromagnetic sector and simplification of the field equations

### FFGFT units (formerly: T0 units)

Setting  $\alpha = 1$  in natural units is standard practice in theoretical physics (Heaviside-Lorentz units, Gaussian units) and not a FFGFT-specific decision. The FFGFT unit system additionally sets  $c = \hbar = \alpha = G = 1$  (Dok. 011; formerly called T0 units).  $\alpha = 1$  is not a posit in the sense of R56 but a *choice of units*: the electromagnetic coupling strength remains the same physical quantity; it receives the numerical value 1. The SI value  $1/137$  does not disappear — it is fully reconstructible from  $\xi$  and  $E_0$ :  $\alpha_{\text{SI}} = \xi \cdot (E_0/1 \text{ MeV})^2 \approx 1/137$  (Dok. 011 **[K]**). In FFGFT units and in SI units  $\alpha$  is the same quantity — only its numerical representation changes.

### Derived electromagnetic quantities

With  $\alpha = 1$  electromagnetic quantities become derived forms of mass and time:

$$\begin{aligned} e &= 1 \quad (\text{charge dimensionless}), \\ V &= E/e = E \quad (\text{voltage} = \text{energy}), \\ I &= e/T = 1/T = m \quad (\text{current} = \text{mass}). \end{aligned}$$

The Coulomb potential simplifies to  $V(r) = -1/r$ .

### Maxwell, Schrödinger and Dirac equations in FFGFT units

The established field equations take their most geometrically transparent form in FFGFT units — not because they are changed, but because the unit choice sets the coupling constants to 1 and thereby makes visible the structure that was always there:

$$\begin{aligned} \text{Maxwell: } \partial_\mu F^{\mu\nu} &= j^\nu, & \partial_{[\mu} F_{\nu\rho]} &= 0, \\ \text{Schrödinger: } i \partial_t \psi &= \left(-\frac{1}{2} \nabla^2 + V\right) \psi \\ \text{Dirac: } (i \not{\partial} - m) \psi &= 0. \end{aligned}$$

$\hbar$ ,  $c$ ,  $e$  and  $4\pi\epsilon_0$  no longer appear explicitly; they are absorbed into the unit choice. This is not new physics but the known physics in a notation that makes the mass structure directly readable:  $i \partial_t \psi = m \psi$  in the free case,  $\partial_\mu F^{\mu\nu} = j^\nu$  without a prefactor.

**What becomes concrete.** Only with Layer 2 does the theory become concrete enough for particle physics: charge quantisation **[B]**,  $\text{su}(3)$  from  $\mathbb{Z}_3$  triality **[B]**, charge definition and gauge structure. The mass layers of the lepton sector are derivable from  $\xi$  **[K]**; the mass ratio  $m_\mu/m_e$  serves as a declared reference point **[POSIT]** (Dok. 190 P42) to keep the layer independently checkable.

### 3 Layer 3: the only free parameter

$\xi = 4/30\,000$  appears as the only dimensionless parameter but is not free given the Galois group orders of the GF tower. From the winding numbers of the  $T^4$  torus (derived from  $\text{char}(\text{GF}(3^n))$ ,  $|\text{GF}(3)^*|$  and the minimal prime divisor of  $|\text{GF}(81)^*|$ ) and the field orders  $|\text{GF}(9)^*| = 8$ ,  $|\text{GF}(27)| = 27$ ,  $5 = \text{min.prime}(80)$ ,  $\xi$  follows algebraically **[K]**:

$$\xi = \frac{(r_\mu/r_e)^2}{8^2 \cdot 5^2 \cdot 27} = \frac{(12/5)^2}{43200} = \frac{4}{30\,000}.$$

(Dok. 336/338; verification scripts `pruef_330`, `pruef_331`.) The premises of this derivation — the winding-number recursion and  $m_\mu/m_e$  as reference point (Dok. 190 P42) — carry a [K] status, not [B].  $\xi$  is therefore not completely free, but not yet completely derived.

With  $\xi$ , Layers 1 and 2 yield measurable scales: Compton wavelength, Hubble length as geometric maximal scale (not an expansion horizon),  $a_0 = cH_0/(2\pi)$  **[K]**.

**What is bracketed.** The cosmic sector  $(H_0, \Lambda)$  is bracketed. The Standard Model derives  $H_0$  equally little; the comparison would be asymmetric (Dok. 190 P39). The exponent 10 in  $H_0 \propto \xi^{10}$  is a posit (Dok. 190 P20); it is structurally motivated but not forced.

**SI anchors are comparison tools.**  $m_e$ ,  $\bar{\lambda}_e$ ,  $v_{CS}$  and all other SI anchors serve exclusively for translation into known unit systems. Every dimensionless statement of FFGFT is a function of  $\xi$  and the posits; no SM value is a structural input. The complexity of comparison tables and precision calculations is communicative, not structural — a concession to established systems that cannot be abandoned without notice.

### 4 Layer 4: algebraic confirmations

The Galois fields  $\text{GF}(9)$ ,  $\text{GF}(27)$ ,  $\text{GF}(729)$ , Hodge theory on  $T^4/\mathbb{Z}_3$ , representation theory of  $\text{Aut}(D_4)$  and comparisons with external carriers (Dok. 361–364) are *consistency checks*, not new inputs. They show that the structure generated by  $\tilde{T} \cdot m = 1$  and the three posits is coherent against known mathematics; they could all have followed from  $\xi$  and the posits alone and in a sense do.

The icosahedron (Dok. 364) is not a special case: every geometric body with non-crystallographic symmetry encounters the same structure — lattice-compatible part in  $\text{Aut}(D_4)$ , remainder as phase in  $\text{GF}(3^k)$ . This is not coincidence but an expression of the prime 3 as fundamental building block: what cannot exist as a root of unity in characteristic 3 must be a lattice operation, and vice versa.

### Tool and method

Claude (Anthropic) was used as a tool for: writing and executing the Python verification scripts; drafting the  $\text{\LaTeX}$  sources from results and corrections of the author; maintaining the changelog and cross-references. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. Every algebraic statement is backed by a Python verification script with explicit assertions — if an assertion fails, the statement is not used.

## 5 Summary: what FFGFT needs and what it does not

| Layer | Content                                           | Status                     |
|-------|---------------------------------------------------|----------------------------|
| 1     | $\tilde{T} \cdot m = 1$ + principle of relativity | fundamental law + 3 posits |
| 2     | $\alpha = 1$ , charge definition                  | posit, EM sector follows   |
| 3     | $\xi = 4/30\,000$                                 | only free parameter        |
| 4     | Galois, Hodge, representation theory              | consistency confirmations  |
| —     | SI anchors, SM comparisons                        | translation, not input     |
| —     | cosmic sector ( $H_0, \Lambda$ )                  | bracketed (SM likewise)    |

What FFGFT does *not* need: the Standard Model, its constants, its Lagrangian as input (only for comparison), external calibrations beyond  $\xi$ . What it delivers: a self-consistent structure that is parameter-lean in natural units and reproduces all known ratios in SI units. It becomes testable only when anchors are inserted — but the anchors are translation, not foundation.

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